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Studierichting: Informatica
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$$\int \frac{17x}{4x^4+15x^2-4}dx = \frac{17}{2}\int \frac{1}{4x^4+15x^2-4}d\left(x^2\right) \overset{x^2=t}{=} \frac{17}{2}\int \frac{1}{4t^2+15t-4}dt$$
$$= \frac{17}{2}\int \frac{1}{(4t-1)(t+4)}dt$$
$$\to \frac{1}{(4t-1)(t+4)} = \frac{A}{4t-1}+\frac{B}{t+4}$$
$$\Rightarrow A = \frac{1}{\frac{1}{4}+4} = \frac{1}{\frac{17}{4}} = \frac{4}{17}$$
$$\Rightarrow B = \frac{1}{4\cdot -4-1} = -\frac{1}{17}$$
$$= \frac{17}{2}\cdot\frac{4}{17}\int \frac{1}{4t-1}dt - \frac{17}{2}\cdot\frac{1}{17}\int \frac{1}{t+4}dt = \frac{1}{2}\int \frac{1}{4t-1}d(4t) - \frac{1}{2}\int\frac{1}{t+4}dt$$
$$= \frac{\ln|4t-1|}{2} - \frac{\ln|t+4|}{2} + c \overset{t=x^2}{=} \frac{\ln|4x^2-1|}{2} - \frac{\ln|x^2+4|}{2} + c$$

References